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How to Improve Coding for Open-Ended Survey Data: Lessons from the ANES

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In surveys, responses to open-ended questions are often released to the public in coded, categorized forms. The typical rationale for converting open-ended responses to coded categories is to protect respondent privacy. Specifically, survey participants sometimes express their responses in ways so unique that others could use that information to identify them. Many leading surveys have concluded that the public release of such information is not worth the risk to respondent privacy and produce codes instead. Many survey researchers, moreover, like the apparent convenience associated with using coded data rather than having to work on their own to translate words into statistically analyzable elements.

Unfortunately, methodological approaches to open-ended coding in survey research have been anything but consistent and credible. The lack of widely accepted best practices might be partially responsible for the decline in the use of the open-ended response format. This is unfortunate because poorly defined methods and best practices should not prevent the use of open-ended responses that can often provide a uniquely rich source of data to survey researchers.

The fundamental question that researchers seeking to code open-ended data must address is “What is the correct inference for a user to draw from a coded open-ended response to a survey question?” The answer to that

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question is going to depend on what question the survey asked, what the respondent says, and then very importantly, on processing decisions that are made after the interview is conducted about converting respondent answers into numbers. A key point is that researchers creating and using these data need to pay greater attention to possible problems caused by inattention to these details.

For this discussion, examples will be drawn heavily from the 2008 ANES, so it is worth outlining the key features relating to open-ended questions on the ANES. There are four general types of questions that the ANES asked in the open-ended format: (1) “most important problem,” (2) candidate “likes-dislikes,” (3) party “likes-dislikes,” and (4) political knowledge.

The “political knowledge” questions are noteworthy because they are among the most frequently used variables that the ANES produces. These questions solicit answers to fact-based quiz questions about politics such as

Now we have a set of questions concerning various public figures. We want to see how much information about them gets out to the public from television, newspapers and the like.... What about...William Rehnquist – What job or political office does he NOW hold?

This question is administered in an open-ended format, allowing respondents to answer in their own words. In most cases, the participant’s complete response has been recorded verbatim. Later, all answers have been coded as either “correct” or “incorrect.” For decades, the publicly available versions of these codes have been treated as valid measures of respondents’ knowledge of the question. However, questions have been raised about the accuracy and relevance of this data.

Generally, ANES data users expect the study investigators to convert open-ended responses into numbers indicating a response as correct or incorrect. A critical point about this process is that the users base their inferences on beliefs about what each of these numbers means. Many users believe that open-ended coding is easy to do, that it generates valid measures, and that it’s performed well by survey organizations. The evidence tells a different story. For the 30 years that the ANES has been asking open-ended recall questions, there is little to no record of users asking critical questions such as: “How did you make decisions about what answers were correct or incorrect?” There is a similarly sparse record of requests for reliability statistics. Yet, the questions are widely used under the assumption that the data are valid and reliable.

Moreover, after examining the ANES's open-ended question data under more intense scrutiny, researchers found that the reality of the state of open-ended coding is much worse than many users assumed. Coding practices for other major surveys were examined they were found to be similarly disappointing. So it is important to evaluate where things have gone wrong.

Consider, for example, problems and controversies associated with the ANES question about the position that William Rehnquist held. Many studies have examined this and similar items and drawn such conclusions as

“Close to a third of Americans can be categorized as ‘know-nothings’ who are almost completely ignorant of relevant political information, which is not, by any means, to suggest that the other two-thirds are well informed.” (Bennett 1998)

“The verdict is stunningly, depressingly clear: most people know very little about politics.” (Luskin 2002)

Indeed, a study by Gibson and Caldiera (2009) found that only 12 percent of respondents provided a “correct” response to the Rehnquist question. At first glance, this statistic appears to support the critics’ conclusions. However, a closer examination of the coding of the responses to the question found that, for the 2004 ANES, responses were marked as “correct” only if respondents specifically identified that Rehnquist was “Chief Justice” and on the “Supreme Court” – meaning that the addition 30 percent of respondents that identified him as a Supreme Court justice were marked “incorrect” due to not specifying “Chief Justice.” When Gibson and Caldiera (2009) asked respondents to state whether William Rehnquist, Lewis F. Powell, or Byron R. White was Chief Justice, 71 percent correctly identified Rehnquist. Similarly, on the 2000 ANES, 400 of the 1,555 respondents said that Rehnquist was a judge or said that he was on the Supreme Court but were coded as having answered incorrectly. Other “incorrect” answers from the 2000 ANES included

- “Supreme Court justice. The main one.”
- “He’s the senior judge on the Supreme Court.”
- “He is the Supreme Court justice in charge.”
- “He’s the head of the Supreme Court.”
- “He’s top man in the Supreme Court.”
- “Supreme Court justice, head.”
- “Supreme Court justice. The head guy.”

- “Head of Supreme Court.”
- “Supreme Court justice head honcho.”

Along similar lines, another political knowledge question asked “...Tony Blair, What job or political office does he NOW hold?” For this item there was a serious error in the coding instructions. ANES coders were told that in order to be marked as “correct,” “the response must be specifically to ‘Great Britain’ or ‘England’ – United Kingdom is *NOT* acceptable (Blair was not the head of Ireland), nor was reference to any other political/geographic unit (e.g. British Isles, Europe, etc.) correct.” In fact, Blair was “Prime Minister of the United Kingdom.” While many scholars and writers used this variable to characterize public ignorance, there is no record that any such user questioned how it was produced prior to subsequent ANES PIs discovering this error.

With this example in hand, we return to questions about how these errors occur and what can be done to improve open-ended coding practice. The answer to the first question is that researchers don’t really know due to poor recordkeeping prior to 2008. Typically for the ANES, interviewers would transcribe the respondent answers and staff would then implement a coding scheme, often with cases coded by a single person. There are few or no records of instructions given to staff regarding coding and there is no documentation of coding reliability analyses. We later discovered that such recordkeeping inadequacies are not particular to the ANES and are common across major surveys.

In response to these discoveries, the ANES moved to make redacted transcripts available whenever possible, conducted a conference to discover and develop best practices, and formed expert committees to develop mutually exclusive and collectively exhaustive coding schemes. The key outcomes included establishing transparent and replicable processes for turning text into numerical data.

The expert discussion about “partial knowledge” was particularly interesting. Recall that the question begins, “What’s the job or political office” that a particular person holds. For example, when this question was asked about Dick Cheney in 2004, a lot of people provided the correct response of “Vice-President.” Others, however, would say “anti-Christ” or “chief puppeteer.” How should these responses be coded? Some experts wanted to count such responses as representing partial knowledge. A breakthrough occurred when the experts returned to the question wording and determined to focus the coding on whether or not a correct answer was given to the question that the ANES actually asked. So if a respondent says that Dick Cheney shot his

friend in a hunting accident, they have decided not to answer the question. They may be providing knowledge about Dick Cheney, but still not answering the question as it was asked. So the whole concept of “partial knowledge” was changed by asking, “is it partial knowledge with respect to this question?”

The ANES’s new coding framework for open-ended questions about political office place emphasis on the following factors:

- Did the respondent say something about the political office?
- Did they identify any part of the title of this person’s political office correctly?
- Some people have multiple offices; for example, the Vice-President of the United States is also President of the Senate. The person who is the Speaker of the House is also a congressperson.

The first factor on which the new coding scheme focuses is “did the respondent say anything correct or partially correct about the political office?” Since the question asks about the person’s job, a correct answer to the question that was asked would also constitute descriptions of what this person does, of what their job is, so things like making legislation or organizing a political party, would also constitute a correct or partially correct response. So for each of the questions the ANES asked, a long list of jobs that a particular person has was identified. So, unlike before where the ANES produced a single “correct” or “incorrect” code for each question, now they have a political office code: “Does the respondent give a correct answer regarding the political office?” The ANES also has a code to indicate if the respondent provided a complete description of a job or an incomplete description? And then finally there is an “other” code. And “other” is anything that the person says that is not pertaining to the job or political office of that person. With respect to the “other” responses, the ANES coding system does not make any judgments about them. In sum, the codes reflect the question, “Did the respondent name the political office, did they name the job, or have they said anything else?”

In terms of procedural transparency, the ANES has decided that it is important for users to be able to see how the code frame was implemented so they can figure out whether there is something about the coding practices that skew the numbers. Written documentation of all decisions, written documentation of all conversations that the ANES had with coding vendors, particularly regarding instructions that confused coders, are now available on

the ANES website. The goal of making such information available is for people who are using the data, or who are developing their own coding schemes, to be able see what the ANES did. With such information in hand scholars who obtain different results than the ANES have better ways of determining “Why?”

Several attributes of the new ANES coding scheme provide some insight into what may be best practices for other surveys or researchers to consider when coding open-ended responses. First, the scheme is theoretically defensible, second, it has demonstrated high inter-coder reliability, third, it is mutually exclusive and collectively exhaustive, and fourth, it is accessible to other scholars who may want to use public data to compare other code frames. More general steps that have been taken include:

- Increased documentation at all stages
- Evaluation at many stages
- Increased procedural transparency
- High inter-coder reliability

In summary, if researchers believe that the scientific credibility of survey-based research depends on transparency and replicability of analysis, then it is imperative that data be manufactured and distributed in ways that facilitate transparency and replicability. The kinds of practices needed to help survey producers and analysts make more effective decisions about how to code open-ended data and interpret the resulting variables are not what they could be. Future research should identify ways to improve practice (and data quality) in ways that need not require extra resources to implement. Doing so should give a wide range of scholars the direction and motivation needed to increase documentation of past coding schemes and engage in more systematic thinking about how best to develop future coding algorithms.

Areas for future research:

- Effective approaches to increasing coding transparency.
- Identifying ways to develop shared coding schemes and practices between survey organizations that are asking the same questions.
- Specifying best practices regarding the production and dissemination of documentation.

References and Further Reading

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